McDonald’s Feedback Analysis
Social Media Command Centre

Problem Statement Reminder
Using GenAI to analyze McDonald's customer feedback and generate actionable insights for management.

Repo link
github.com/IS492-SP26/team-project-social-media-command-centre/

Checkpoint #2 Presentation | IS492 | University of Illinois Urbana Champaign
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What We Needed to Validate

Assumption	Initial Belief	Why It Could Be False	Mitigating Feature
Reviews are representative	Reflects all customers	Skews toward extreme opinions	User survey validation (15 participants)
RoBERTa fits restaurant text	Tweet-trained model works on reviews	Language patterns differ significantly	RoBERTa pre-trained on 124M tweets
Sentiment = actionable insight	Scores alone guide operations	Lacks specificity on issue type	Aspect-level sentiment tagging + geographic drill-down

What’s New and What’s Next ?

Checkpoint 1
Explore problem scope and feasibility

Checkpoint 2
Extract, clean, transform, visualize data (because unclean data == useless data)

Checkpoint 3
Develop and validate the NLP GenAI model

Checkpoint 4
Showcase PowerBI dashboard and analyze user adoption

Prompting Study Design

Tools Tested: ChatGPT, Claude, Gemini

Why? To check if generic GenAI NLP models work for our purpose

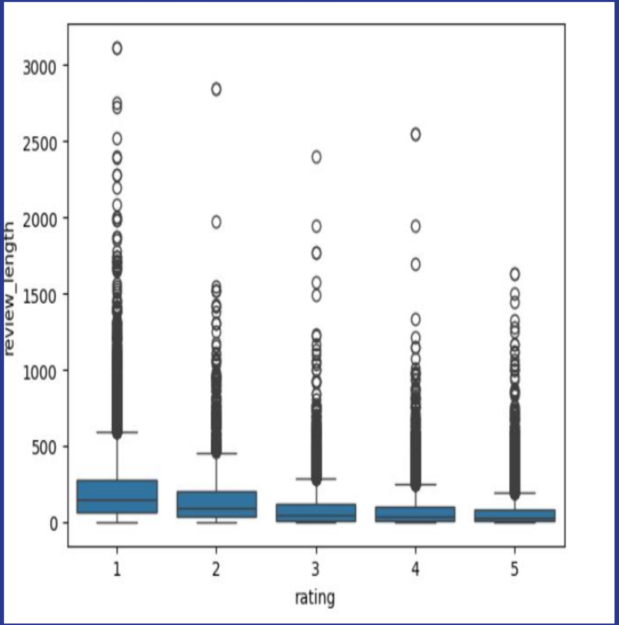
Task Types

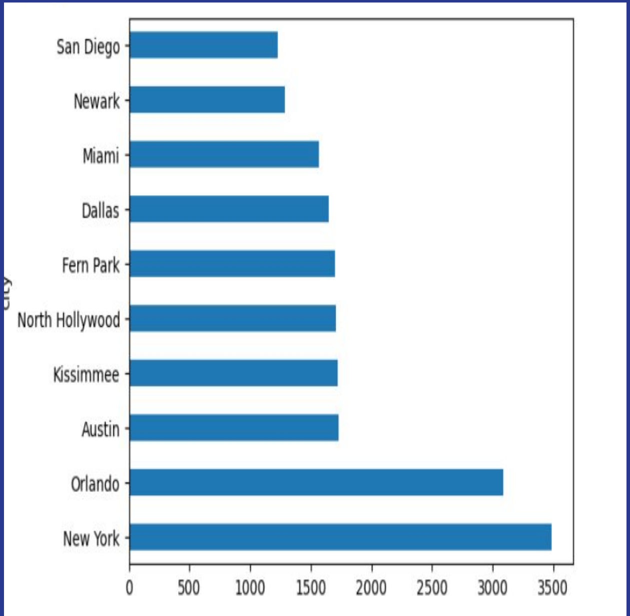
- **Typical**: Sentiment classification on 10 real reviews
- **Edge Case:** Sarcasm detection ("Great job, I waited 25 mins for cold fries")
- **Failure Case:** Ambiguous review — confidence score ~45%

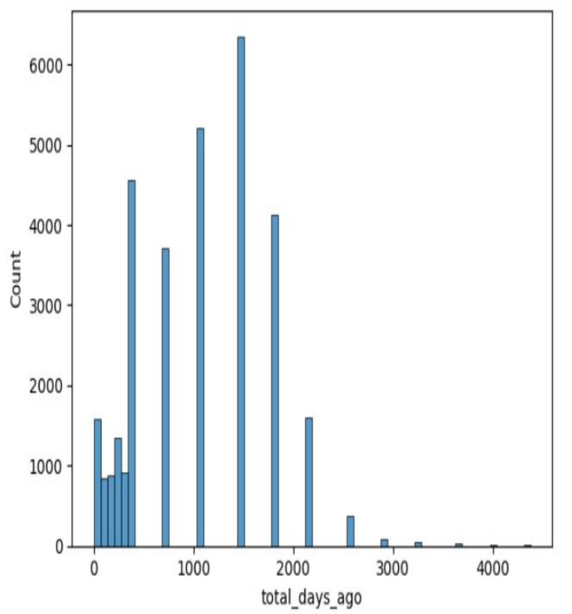
Prompt Control: All 3 models received identical prompts, inputs, and output format instructions

Conclusion: Specific Tailored models Needed (more on that later)

Evidence: Data Behaviour + Credibility







Gap Analysis & Opportunity

Existing tools rely on sentiment & star ratings → miss **operational context**

Fail on sarcasm, mixed sentiment, vague feedback → risk **misclassification**

Rating-text mismatches hide real issues → **undetected problems**

No uncertainty awareness → **ambiguous reviews ignored**

Tool	Limitation	Our Opportunity
Sentiment dashboards	Only pos/neutral/neg	Extract operational issues
Keyword-based tools	Misinterpret language	Context-aware LLM understanding
Rating-based systems	Rating-text mismatch	Detect hidden issues
Traditional ML	No uncertainty	Flag ambiguous reviews

Risks, Safeguards & Path to CP3

Risk Type	Risk	Mitigation
Privacy	Reviews contain real names or locations	Strip PII during data cleaning
Reliability	Model misclassified sarcasm or ambiguity	Flag low confidence outputs below 60 percent for human review
Cost	LLM API calls at scale are expensive	Use batch processing and cache repeated queries
Reliability	Hallucinated operational recommendations	Ground outputs strictly in actual review text

Checkpoint 3 Target: Setup the NLP model framework

Questions ?

Constructive Feedback Is Welcomed !